AP PRECALCULUS

Unit 1 • Lesson 1.12 notes

Lesson 1.12 • Transformations of Functions

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Describe translations, dilations and reflections of a graph.
- Write g in terms of f from a description or a graph.
- Find the domain and range of a transformed function.

Vocabulary. translation, vertical dilation, horizontal dilation, reflection, domain, range.

1. The four parameters

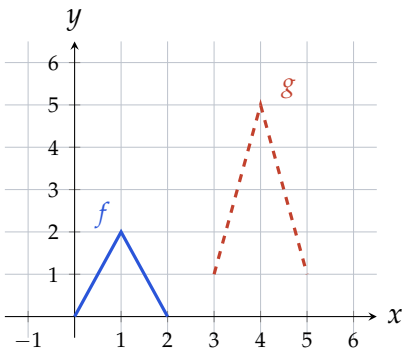
Key idea

In $g(x) = a\,f\big(b(x-h)\big)+k$:

- k shifts the graph up ($k>0$) or down; h shifts it right ($h>0$) or left.
- a stretches vertically by $|a|$; $a<0$ also reflects over the x-axis.
- b stretches horizontally by $\frac{1}{|b|}$; $b<0$ also reflects over the y-axis.

A point (x,y) on f moves to $\left(\frac{x}{b}+h,\ ay+k\right)$ on g.

Example 1 • From a graph



Solution. The peak $(1,2)$ moves to $(4,5)$ and the base points $(0,0),(2,0)$ to $(3,1),(5,1)$. The width is the same, the height doubles ($2\to 4$ above the base), the graph moves right 3 and up 1: $g(x)=2f(x-3)+1$. Check: $2f(4-3)+1=2(2)+1=5$.

Example 2 • Domain and range

f has domain $[-4,2]$ and range $[-1,3]$. Find the domain and range of $g(x)=-f(2x)+1$.

Solution. Domain: $2x\in[-4,2]$, so $x\in[-2,1]$. Range: $-f$ takes values in $[-3,1]$, then $+1$: $[-2,2]$.

Common mistakes

- Shifting the wrong way: $f(x+3)$ moves the graph *left* 3.
- Thinking $f(2x)$ stretches the graph horizontally; it compresses it by a factor of $\frac{1}{2}$.

Quick check

1. $(2,5)$ is on f. Which point is on $y = f(x+1) - 3$? 2. Describe $y = -\frac{1}{2}f(x)$.

Answers. 1. $(1,2)$. 2. Vertical dilation by $\frac{1}{2}$ and reflection over the x-axis.